

Pratyush Das

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Education

Purdue University <i>PhD in Computer Science (Advisors - Milind Kulkarni, Ben Delaware)</i> Research interests - Compilers, Automatic parallelization, Sparse tensors, High performance computing	August, 2021 - May, 2027 (Expected)
Institute of Engineering & Management, Kolkata (MAKAUT) <i>Bachelor of Technology in Computer Science and Engineering</i> Awarded the Director's Award for Best Scientific Mind	August, 2017 - May, 2021

Experience

ML Performance (Google Cloud) - Student Researcher Google • XLA TPU compiler: Built out an XLA plugin estimating TPU cycles of HLO ops without actually running on TPUs - Broadened HLO op coverage (e.g. all-reduce) and added support for more LLO emitters (pipeline, collectives). - Reduced delta against measured TPU runs on Goose, Gemini, and YouTube workloads from over 50% to ~5%. - Sped up static estimation by implementing compiler passes such as loop fast-forwarding.	May, 2026 - August, 2026
SwiftUI - Compiler Intern Apple • Swift compiler: Extended the compiler and runtime to retarget SwiftUI to a new IR - Extended Swift's code generation and runtime, with LLVM patches, to lower SwiftUI to a new target IR. - Integrated an interpreter into Swift's runtime, enabling live updates of SwiftUI views from newly loaded code. - Implemented dynamic loading and symbol resolution of system shared libraries for interpreted code.	May, 2023 - August, 2023
Google Summer of Code - Student The LLVM Compiler Infrastructure Organization Supervisors - William Moses, Johannes Doerfert • Enzyme: LLVM Pass to perform automatic differentiation of statically analyzable LLVM IR - Integrated custom derivatives of several BLAS functions into Enzyme. - Wrote an LLVM pass to inline function definitions from bitcode files into LLVM IR.	June, 2021 - August, 2021
IRIS-HEP - Fellow Princeton University Supervisor - Jim Pivarski • Awkward Array: Library for nested, variable-sized data using NumPy-like idioms - Implemented a source-to-source compiler from a C++ subset to equivalent Python, with property-based testing. - Created a source-to-source compiler to generate equivalent parallel CUDA from specification (Python and type info). - Enabled automatic parallelization of user-written Awkward Array kernels.	June, 2020 - September, 2020
IRIS-HEP and DIANA-HEP - Fellow Fermi National Accelerator Laboratory and Princeton University Supervisor - Jim Pivarski • Uproot: Python implementation of ROOT I/O, an open source file format storing over an exabyte of HEP data - Enabled writing fundamental HEP data structures like TTrees and histograms to ROOT files. - Uproot has become one of the most widely used HEP libraries.	June, 2018 - September, 2018; June, 2019 - September, 2019

Publications

- **P.Das**, A.Xhebraj, T.Rompf, "Specializing Data Access in a Distributed File System (Generative Pearl)", 23rd ACM SIGPLAN International Conference on Generative Programming: Concepts and Experiences (GPCE, 2024), DOI: <https://doi.org/10.1145/3689484.3690736>.
- T.Mustafa, K.Kallas, **P.Das**, N.Vasilakis, "DiSh: Dynamic Shell-Script Distribution", 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI, 2023).
- J.Pivarski, I.Osborne, **P.Das**, D.Lange, P.Elmer, "AwkwardForth: accelerating Uproot with an internal DSL", 25th International Conference on Computing in High-Energy and Nuclear Physics (vCHEP, 2021), DOI: 10.1051/epj-conf/202125103002.
- J.Pivarski, I.Osborne, **P.Das**, A.Biswas, P.Elmer, "Awkward Array: JSON-like data, NumPy-like idioms", Proceedings of the 19th Python in Science Conference (SciPy USA, 2020), Pages 68-74, DOI: 10.25080/Majora-342d178e-00b.
- E.Rodrigues, et al., "The Scikit HEP Project - overview and prospects", Proceedings of the 24th International Conference on Computing in High Energy and Nuclear Physics (CHEP, 2019), DOI: 10.1051/epjconf/202024506028.

- N.Saha, **P.Das**, H.N.Saha, “Authorship Attribution of Short Texts using a Multi Layer Perceptron”, International Journal of Applied Pattern Recognition, 2018 Vol. 5 No. 3, Pages 251-259, DOI: 10.1504/IJAPR.2018.10016100.

Preprints under submission

- **P.Das**, A.Basareh, A.Pelenitsyn, K.Sundararajah, M.Kulkarni, B.Delaware, “Bring Your Own Formats and Kernels: Composable Abstractions for Sparse Matrix Computation”.
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Other Open Source Contributions

Supervisor - Vassil Vassilev

November, 2019 - May, 2021

- **ROOT**: An open-source data analysis framework storing over an exabyte of data
 - Improvements to interpreter (rootcling)
- **Cling**: Interactive C++ interpreter built on top of Clang
 - Maintained cpt.py installer and packager
- **Clang**: C language family frontend for LLVM
 - Several patches to print type information of C++ template arguments

Supervisors - Jim Pivarski, Viktor Khristenko

June, 2017 - August, 2017

- **spark-root** - Apache Spark datasource for ROOT
 - Separated spark bindings from TTree reading code
 - **root4j** - Java implementation of ROOT file reader
 - Optimized codebase to facilitate interoperability
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Teaching Experience (Teaching Assistant)

CS 307: Software Engineering - Purdue University

Spring 2025, Fall 2025

CS 354: Operating Systems - Purdue University

Fall 2022 - Fall 2024, Spring 2026

CS 240: Programming in C - Purdue University

Fall 2021, Spring 2022

Programming Languages and Tools

Experienced: Python, CUDA, XLA

Familiar: C, C++, Java, Scala, Coq, Bash, L^AT_EX, Swift, WebAssembly, MLIR

Summer Schools

Oregon Programming Languages Summer School - University of Oregon

2021

Computational and Data Science for High Energy Physics - Princeton University

2019

Talks at Conferences

- Specializing Data Access in a Distributed File System (GPCE) 2024
 - GSoC Experience - Enzyme (LLVM Developers' Meeting) 2021
 - Python in High Energy Physics (SciPy India, PyCon USA) 2019, 2020
 - Writing files with uproot (PyHEP) 2019
 - Writing files with uproot (ROOT Users' Workshop) 2018
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Invited talks at Research Groups

- Staging Blocked Evaluation of Sparse Matrix Computations (PurPL Seminar) 2026
 - Language Transformations for the Awkward Array library (IRIS-HEP Fellow Presentations) 2020
 - CUDA backend for the Awkward Array project (Princeton University Liberty Research Group) 2020
 - Writing TTrees with uproot (IRIS-HEP Topical Meeting: Summer student project presentations) 2019
 - Writing files with uproot (DIANA Meeting: Updates on ROOT I/O) 2018
 - Separation of Concerns - Refactoring code between ROOT4J and Spark-Root (CMS Big Data Science, DIANA-HEP) 2017
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Posters

- SABLE: Staging Blocked Evaluation of Sparse Matrix Computations (Midwest PL Summit) 2024
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Service

- Purdue Programming Languages group (PurPL) co-ordinator 2024,2025
- Artifact Evaluation Committee (SOSP) 2023